Sampling Methods

Questions 1–16

Question 1. A by-election is due to take place in Wellingborough. Amelia is a politics student who is interested in the election and decides to survey local residents. She stands in the main high street and surveys the first 500 people she meets. Identify the name of this type of sampling.

simple random opportunity
cluster quota

Question 2. In a small village of 156 households, Jason wants to investigate how many use the local post office. He decides to number the households from 001 to 156, and then use a random number generator to obtain 30 different three-digit numbers between 001 and 156. He will then select the corresponding households and include them in the sample. Identify the name of this type of sampling.

convenience self-selection
simple random systematic

Question 3. The table below shows the share price of HSBC on the 1st January, 1st April, 1st July and 1st October across a sample of five years.

	01/01	01/04	01/07	01/10
1993	156.57	173.13	189.10	221.34
1998	464.77	548.13	442.40	406.67
2003	549.88	597.37	671.01	771.22
2008	654.88	766.43	731.13	664.03
2013	716.70	703.50	747.80	682.10

Identify the sampling procedure that most likely chose the years selected in the sample.

quota systematic
simple random stratified

Question 4. In a secondary school, there are 12 classes in year 9 and the school wants to collect some feedback on a recent futures fair that visited the school. Sandra, the head of year 9, decides to select three classes at random and then get feedback from every student in those classes. Identify the name of this type of sampling.

stratified quota
opportunity cluster

Question 5. Ibrahim works for a local charity and is asked to interview members of the public in the high street on a Saturday afternoon. He is asked to interview 30 people under the age of 40, and 30 people who are aged 40 or over. Identify the name of this type of sampling.

simple random stratified
systematic quota

Question 6. A large retailer of flat-pack furniture employs 7200 members of staff across 24 stores in the UK. The retailer wants to investigate their staff's job satisfaction through a survey. The HR team suggest four possible sampling methods:

Method A Randomly select 120 people from the whole list.
Method B Randomly select five stores and then randomly select 24 people from each of these stores.
Method C Randomly select 120 people from the London store.

a) Identify the population in the retailer's investigation.
b) Give one advantage of using Method B over Method C.
c) Give one advantage of using Method B over Method A.

Question 7. The Office for National Statistics conducts a census for England and Wales every 10 years. Describe one disadvantage with a census.

Question 8. In 2019, there were estimated to be approximately 20,850 breeding pairs of African penguins. The health of the population is to be investigated by sampling 100 breeding pairs. Explain why a simple random sample cannot be used.

Question 9. A national union of care workers is holding its annual conference. Chloe wants to sample the views of care workers on conditions in their place of work. She decides to set up a stall near the entrance to the conference and ask attendees who visit her stall to fill in a questionnaire. State two disadvantages with this sampling method.

Question 10. Clive runs a takeaway business. He wants to revise the menu down to reduce costs. He has recorded the choices made by each customer since the business opened and now has 680 customers and their choices in a spreadsheet. Describe how Clive could obtain a simple random sample of size 40 from his list of customers.

Question 11. Lundi supermarkets split their workforce into those who work on the shop floor and those who work in the warehouse. Across all of their stores, 3265 work in the shops and 2435 work in the warehouses. The company wants to conduct a survey of 150 members of staff, and they decide to use a stratified sample. Calculate how many from the shop floor and how many from the warehouses should be included in the sample.

Question 12. Harry works in a large office block. Near his office, there is a kitchen where employees store their mug in a cupboard. The mugs are all different, but his is often used by somebody else accidentally. Harry wants to label all the mugs with their owner's name. Suggest why an opportunity sample conducted near the kitchen would be more appropriate to gather opinions than a simple random sample to the whole workforce.

Question 13. Susan is the Subject Leader of Maths at a large sixth form college. She wants to investigate the number of hours it takes for the students taking A-Level Maths to complete their homework each week. There are 1200 students in 50 classes. Describe how Susan could obtain a cluster sample of three classes of students for her survey.

Question 14. A company wishes to survey its employees. It decides to send out an online survey to all of their employees' work emails and then analyse the results they get within a 24-hour window. Suggest two disadvantages with using this method.

Question 15. A work-place choir is made up of four companies: Aurora, BrightByte, Cosmic and Dreamscape. The number of people from each company who take part in the choir are 25, 30, 50 and 45 respectively. A stratified sample is taken of size k to gain feedback on the choir, and the samples taken from Aurora, BrightByte and Cosmic are 8, 10 and 17 respectively. In the calculations to determine how many people must be sampled from each company, the number was always rounded to the nearest person.

a) Determine the value of k

b) Determine how many people who work at Dreamscape should be sampled.

Question 16. A sixth form college employs 120 members of staff. A survey is to be sent out to 20 members of staff using the following method:

- List all of the members of staff in alphabetical order by surname and assign each of them a number from 1 to 120.
- Choose a random number from 1 to 6 and choose this person.
- Select every 6th person on the list from there until 20 people are selected.

a) Identify this sampling method.

b) Give a reason why this method does not provide a random sample.